# Constructive Solvability of the Level 3 Menger Twist Cube

Menger Cube Project

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**Abstract**

The Level 3 Menger Twist Cube has 8,000 labeled cells. We show that the `level3-slice-reduction` implementation returns every state reachable under its 3,591 legal atomic moves to the solved state in finitely many moves. The proof combines orbit decomposition, alternating-group actions, orbit-local 3-cycles, parity normalization, and orientation invariants. We characterize the reachable states and determine the order of the move group.

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# 1 Model

A Level $n$ grid has side length $3^n$. Write each position uniquely in balanced ternary as

$$p = \sum_{j=0}^{n-1} 3^j d_j, \quad d_j \in \{-1, 0, 1\}^3, \qquad P_n = \{p : z(d_j) \le 1 \text{ for every } j\},$$

where $z(d)$ counts the zero coordinates. Thus $|P_n| = 20^n$. At Level 3 write $p = 9B + 3b + o$. Let $C$ be the 24-element rotation group of the cube. A state $s = (\sigma, \rho)$ assigns each cell, labeled by its home $x \in P_3$, a position $\sigma(x)$ and an orientation $\rho(x) \in C$. The solved state has $\sigma = \mathrm{id}$ and $\rho(x) = I$.

An atomic move specifies a target $D \subseteq P_n$, an axis $a$, a point $c$ on the axis, and an angle $\theta \in \{90°, -90°, 180°\}$. On selected cells it sends $p$ to $c + R_{a,\theta}(p - c)$ and multiplies orientations on the left by $R_{a,\theta}$. Cells are closed unit cubes. The collision test permits boundary contact and excludes positive-volume overlap between moving and stationary cells during the rotation.

Table 1: Level 3 targets in the standard move system.

| Family | Candidates | Legal | Description |
|---|---|---|---|
| Frame (scale 1) | 81 | 81 | One-cell-thick slice |
| Frame (scale 3) | 27 | 27 | Three-cell-thick slice |
| Frame (scale 9) | 9 | 9 | Nine-cell-thick slice |
| Extension (depth 1) | 12 | 12 | Outermost edge block |
| Slab (depth 1.5) | 36 | 36 | Slab of an outer block |
| Extension (depth 2) | 240 | 72 | Middle block |
| Slab (depth 2.5) | 720 | 216 | Slab of a middle block |
| Extension (depth 3) | 4800 | 744 | Single-cell turn |
| Total | 5925 | 1197 | |

The three angles for each legal target give $|S_3| = 1197 \cdot 3 = 3591$ atomic moves. Let $G_3 = \langle S_3 \rangle$. Reachability means reachability from the solved state by a word in $S_3$. We apply words from left to right. For example, $uTu^{-1}$ transports the support of $T$ back through $u$. A commutator $ABA^{-1}B^{-1}$ is even because the sign map is multiplicative.

The scope is the standard generator set. For example, rotating the four cells $(\pm 5, \pm 5, 0)$ about the $z$ axis through the origin passes the collision test but is outside $S_3$.

## 2 Main results

**Theorem 2.1** (Constructive solvability)**.** *For every reachable Level 3 state,* `level3-slice-reduction` *constructs a finite word of legal atomic moves that solves it, provided that the computer has sufficient resources for the finite searches.*

**Theorem 2.2** (Reachability and group order)**.** *A state is reachable if and only if it meets all four conditions below. The orientation coordinates and maps are defined in Section 5.*

1. *The position permutation preserves every orbit $\Omega$, and $\tau_s(x) \in H(\Omega)$ for each $x \in \Omega$.*
2. *The orbit parity vector $\varepsilon(\sigma)$ lies in the 15-dimensional subspace $V \subseteq \mathbb{F}_2^{164}$ spanned by the atomic move parity vectors.*
3. *On each of the four $C_3$ orbits, $\sum_{x\in\Omega} \chi(\tau_s(x)) = 0$ in $\mathbb{Z}/3$.*
4. *On the single $D_4$ orbit, $\sum_{x\in\Omega} \varphi(\tau_s(x)) = 0$ in $\mathbb{Z}/2$.*

*The position image is $P = \{\pi \in \prod_i \mathrm{Sym}(\Omega_i) : \varepsilon(\pi) \in V\}$, and*

$$|G_3| = \left(2^{15} \prod_{i=1}^{164} \frac{|\Omega_i|!}{2}\right) 3^{28} 4^{552} \frac{8^{192}}{2}, \qquad \log_{10} |G_3| \approx 11918.138.$$

## 3 Position orbits and local generators

Connect $x$ to $a(x)$ for every $a \in S_3$. The connected components are the position orbits: edges can be followed using generators or their inverses, and every word traces such a path. Enumerating the atomic actions gives the following decomposition.

| Orbit size | Orbits | Cells |
|---:|---:|---:|
| 8 | 4 | 32 |
| 24 | 112 | 2688 |
| 96 | 45 | 4320 |
| 192 | 1 | 192 |
| 384 | 2 | 768 |
| Total | 164 | 8000 |

Each of $B, b, o$ is of type C (no zero coordinate) or E (one zero coordinate). The coincidence pattern of the E axes refines these types. Coordinate rotations preserve the resulting 15 classes.

| Class | Cells | Orbits | Size by multiplicity |
|---|---|---|---|
| CCC | 512 | 24 | $8 \times 4,\ 24 \times 20$ |
| CCE | 768 | 32 | $24 \times 32$ |
| CEC | 768 | 26 | $24 \times 24,\ 96 \times 2$ |
| CEE/$bo$ | 384 | 13 | $24 \times 12,\ 96 \times 1$ |
| CEE/$b\|o$ | 768 | 26 | $24 \times 24,\ 96 \times 2$ |
| ECC | 768 | 8 | $96 \times 8$ |
| ECE/$Bo$ | 384 | 4 | $96 \times 4$ |
| ECE/$B\|o$ | 768 | 8 | $96 \times 8$ |
| EEC/$Bb$ | 384 | 1 | $384 \times 1$ |
| EEC/$B\|b$ | 768 | 8 | $96 \times 8$ |
| EEE/$Bbo$ | 192 | 1 | $192 \times 1$ |
| EEE/$Bb\|o$ | 384 | 1 | $384 \times 1$ |
| EEE/$Bo\|b$ | 384 | 4 | $96 \times 4$ |
| EEE/$B\|bo$ | 384 | 4 | $96 \times 4$ |
| EEE/$B\|b\|o$ | 384 | 4 | $96 \times 4$ |

The catalogue lists individual orbits in `wiki/Level-3-Orbit-Catalogue.md`.

**Lemma 3.1.** *The induced position group $G_3^\Omega$ contains* $\mathrm{Alt}(\Omega)$ *on every orbit* $\Omega$*.*

*Computer-assisted proof.* For the four size-eight orbits, closure enumeration yields $8! = 40320$ elements. On each larger orbit, verify primitivity by closing the smallest generator-invariant equivalence relation identifying each pair of distinct points. Each closure has one class. Then verify a word $g$ and an exponent $k$ such that $g^k$ acts as a single prime-length $p$-cycle with $p \leq |\Omega|-3$. Jordan's theorem gives the conclusion. Randomness is used to find candidate words. The final check evaluates the actual permutations. □

Call a word pure on an orbit if it fixes positions and orientations outside the orbit and leaves orientations unchanged at every site it fixes. The stored nontrivial even tools have support sizes 3, 4, and 6 on 133, 17, and 14 orbits respectively. Of these, 117 were obtained by combining small-support commutators and 47 by commutators with conjugates. The words have lengths 16–84, with median 16. All 164 passed an audit of move legality, support, and orientation at fixed sites.

**Lemma 3.2** (Pure 3-cycles)**.** *Every orbit has a pure 3-cycle given by an explicit legal word. Every ordered triple of distinct sites on that orbit can be cycled by a pure word.*

*Proof.* The stored size-three tools cover 133 orbits. On the other 31, the original tool is a double transposition or a product of two 3-cycles. Breadth-first

search over setup words found two conjugates whose product has support exactly three. The resulting 42–170 move words are stored in `level3SliceReductionCertifiedData.ts`. Check C5 evaluates all 164 words on all 8,000 positions and orientations.

By Lemma 3.1 and $|\Omega| \geq 8$, the induced group acts transitively on ordered triples of distinct sites. Choose $u$ sending the desired triple to the stored one, apply the stored cycle $T$, and undo $u$. Outside the three chosen sites, $T$ acts trivially after transport, and $u^{-1}$ restores both position and orientation. □

Every even permutation is a product of 3-cycles: shorten long cycles using triples sharing their first point and express each remaining pair of transpositions as at most two 3-cycles. Lemma 3.2 therefore realizes any even permutation of one orbit without disturbing other orbits. Containment of $\mathrm{Alt}(\Omega)$ in the induced group alone gives no such independence.

# 4 Parity normalization and position solving

The signs on the 164 orbits define a homomorphism $\varepsilon : G_3 \to \mathbb{F}_2^{164}$, with bit one for odd sign. The atomic move vectors give 20 distinct nonzero vectors of rank 15. Gaussian elimination selects at most 15 atomic moves whose vectors sum to the input parity. Applying their inverses makes each orbit even.

The solver processes one orbit at a time. While more than 32 sites remain misplaced, it uses a breadth-first search on ordered pairs to find a conjugate of a pure 3-cycle that increases the number of correct sites. It switches to the exact phase when no improvement is found. The number of correct sites bounds the number of greedy steps. In the exact phase, the solver decomposes the remaining even permutation into 3-cycles. It constructs each required conjugate by bidirectional breadth-first search on ordered triples, from the requested triple and the stored one. Lemma 3.2 supplies a path, and exhaustive search of this finite graph terminates. Each shortening step solves at least two sites, or four sites when removing a pair of disjoint transpositions. Every completed orbit stays solved in position.

# 5 Orientation

## 5.1 Local coordinates

Fix a base site $x_\Omega$ in each orbit. Let $Q_y$ be the set of orientations the base cell can have at $y$ and put $H(\Omega) = Q_{x_\Omega}$. The latter is a subgroup of $C$ by composition and inversion of loops. Choose $r_y \in Q_y$. Following a reference path to $y$ after a loop gives $r_y H(\Omega)$. Returning along the reference path shows the reverse inclusion, so $Q_y = r_y H(\Omega)$. Exhaustive search on $\Omega \times C$ computes these sets.

| $H(\Omega)$ | Orbits | Cells | Single-cell turns | Constraint |
|---|---|---|---|---|
| Trivial | 148 | 7224 | None | Trivial orientation |
| $C_3$ | 4 | 32 | None | Zero twist sum |
| $C_4$ | 11 | 552 | Every site | None |
| $D_4$ | 1 | 192 | Rotation subgroup | Even flip count |

Write $\pi_a$ for the position action and $R_a(p)$ for the applied rotation at $p$ (identity when $p$ is unselected). Define

$$\tau_s(x) = r_{\sigma(x)}^{-1}\rho(x)r_x, \qquad \delta(a,p) = r_{\pi_a(p)}^{-1}R_a(p)r_p.$$

Since $R_a(p)r_p \in Q_{\pi_a(p)}$, we have $\delta(a,p) \in H(\Omega)$. Substitution gives $\tau_{s\cdot a}(x) = \delta(a,\sigma(x))\tau_s(x)$. Starting from the solved state gives $\tau_s(x) \in H(\Omega)$ for every reachable state. For a word $g$, multiply the successive increments in reverse chronological order to obtain $\delta(g,p)$. In particular $\delta(g^{-1},\pi_g(p)) = \delta(g,p)^{-1}$. If $g$ fixes every position, applying $u$, then $g$, then $u^{-1}$ gives

$$\delta(ugu^{-1},p) = \delta(u,p)^{-1}\delta(g,\pi_u(p))\delta(u,p).$$

Thus transport conjugates the increment within $H(\Omega)$.

### 5.2 Invariants and orientation tools

On a $C_3$ orbit fix an isomorphism $\chi : H(\Omega) \to \mathbb{Z}/3$. On the $D_4$ orbit let $C(\Omega)$ be the order-four rotation subgroup and let $\varphi : H(\Omega) \to H(\Omega)/C(\Omega) \cong \mathbb{Z}/2$ be the quotient map. For each atomic move, Check C6 verifies that the sum of its local increments under $\chi$ or $\varphi$, as appropriate, is zero. Because the maps are homomorphisms and $\sigma$ permutes each orbit, applying a move changes the sum by exactly the sum of its increments. Thus the sums are invariant and vanish in reachable states.

Check C7 verifies that on all 744 sites in the $C_4$ and $D_4$ orbits, single-cell turns realize all three nonidentity rotations in the local rotation subgroup. Check C8 verifies these additional words:

- On each $C_3$ orbit a 68-move word $P_\Omega$ fixes every position and adds opposite nonzero twists at two sites, leaving all other orientations fixed. It is obtained as $TuT^{-1}u^{-1}$, where $T$ is a pure cycle and $u$ cyclically permutes its support.
- On the $D_4$ orbit a 32-move word $F$ fixes every position and flips two orientations while leaving the rest fixed. This word is one of 13 previously stored $D_4$ tools.

### 5.3 The orientation kernel

Let $K$ be the kernel of the position action. The local orientation condition and the sums above constrain $K$ to the product, over orbits, of the zero-sum $C_3$ vectors, all $C_4$ vectors, the even-flip $D_4$ vectors, and the trivial factors.

For the reverse inclusion, single-cell turns generate the $C_4$ factors. On a $C_3$ orbit, the induced group is transitive on ordered pairs by Lemma 3.1. Transport $P_\Omega$ to any ordered pair. Conjugation preserves its increments since $C_3$ is abelian. Such transfers generate every zero-sum vector. On the $D_4$ orbit, transport $F$ to any pair. Conjugation preserves its flip bits because $\varphi$ has abelian image. Pair the nonzero flip bits, cancel them, and use single-cell rotations for the residual orientations. Each of these words acts within its orbit, so the factors can be generated independently. Consequently

$$|K| = 3^{32-4}4^{552}\frac{8^{192}}{2}.$$

*Proof of Theorem 2.2.* Necessity follows from the orbit decomposition, parity homomorphism, local coordinate condition, and two invariant sums. For sufficiency, choose a legal word whose parity equals that of the input. The residual position permutation is even on each orbit, so Lemma 3.2 realizes it independently on every orbit. Together these words give a group element with the same position permutation as the input. Applying its inverse to the input fixes the positions and preserves the orientation conditions. By the kernel description the residual state lies in $K$, so the input is reachable.

The same position argument identifies $P$. Each parity vector in $V$ has a fiber of size $\prod_i |\operatorname{Alt}(\Omega_i)|$, giving the stated order of $P$. Multiplying by $|K|$ gives the order of $G_3$. □

### 5.4 Constructive orientation correction

Once positions are solved, the heuristic phase searches conjugates of position-fixing tools, including single-cell turns, differences of tools with identical position action, commutators, and stored $D_4$ words. It accepts only moves that strictly decrease the sum of orientation distances to the solved state. This nonnegative integer potential ensures termination, even if the heuristic stalls.

The completion phase first applies available single-cell corrections. This solves the $C_4$ orbits and leaves only flipped sites on the $D_4$ orbit. On each $C_3$ orbit, choose two twisted sites and use breadth-first search on ordered pairs to transport $P_\Omega$ to them. One or two applications solve the first site, and the number of twisted sites strictly decreases. The zero-sum constraint excludes a final lone twist. On the $D_4$ orbit, pair the flipped sites and transport $F$ to cancel each pair of flips. Single-cell rotations then solve the pair. The number of flipped sites decreases by two each time. All these

words fix positions. The implementation reports violations of the parity or orientation sum conditions when encountered.

*Proof of Theorem 2.1.* Normalize orbit parity, solve positions orbit by orbit, and solve the remaining orientations by the terminating completion procedure. Every phase outputs legal atomic moves, so their concatenation solves the input. □

## 6 Replay and move counts

Search uses a sparse integer simulator. The solver replays its output in the puzzle engine from a copy of the input, checks legality before each move, and verifies the final positions and orientations of all 8,000 cells. Any mismatch or illegal move fails replay. A separate comparison checked the sparse simulator against the engine over six trials of 40 legal moves, checking all 8,000 cells.

Seven runs with the earlier implementation produced solutions of 111,830–424,519 moves in 112.1–284.2 seconds. For the worst-case shortest solution length, the number of words of length at most $d$ is

$$1 + M + \cdots + M^d = \frac{M^{d+1} - 1}{M - 1}, \qquad M = 3591.$$

If this count is smaller than the number of states being counted, some state requires more than $d$ moves.

| States counted | $\log_{10}$ of count | Lower bound |
|---|---|---|
| $\prod_i \mathrm{Alt}(\Omega_i)$ (earlier bound) | 11394.834 | 3206 |
| Position group $P$ | 11399.349 | 3207 |
| Full group $G_3$ | 11918.138 | 3353 |

Thus God's number in the standard atomic-move metric is at least 3,353. The measured solver outputs are individual solutions of unknown optimality.

## 7 Verification and reproduction

- `npmrunprove:l3` runs checks C1–C9 deterministically via `research/level3-completeness/certify.ts` and writes `research/level3-completeness/out/certificate.json`.
- `npmruntest:l3` also solves directly constructed states satisfying the four reachability conditions, in both normal mode and with the heuristic phases disabled. It tests rejection of states violating each orientation sum separately.

- `npmrunverify:l3` audits stored tools, compares simulators, and checks the orbit group certificates. The corresponding scripts under `research/scratch/` are `l3-audit-tooldata.ts`, `l3-sim-vs-engine.ts`, and `l3-orbit-groups.ts`.
- To reproduce a solve, run the following command:

  ```
  npm run bench -- --algorithm=level3-slice-reduction
    --level=3 --seeds=6 --length=100
  ```

  Join the two lines into one command. The run takes several minutes.
- The solver and tool data are under `packages/solver-core/src/algorithms/`. `level3SliceReductionSolver.ts` implements the solver. `level3SliceReductionToolData.ts` contains the original pure tools and $D_4$ words. `level3SliceReductionCertifiedData.ts` contains the additional 31 pure cycles and orientation tools, generated by `research/level3-completeness/find-tools.ts`.

**Check C1.** There are 3,591 atomic moves, closed under inversion.

**Check C2.** The 164 position orbits have the listed sizes.

**Check C3.** Exhaust the position-orientation automata and verify the cosets $Q_y = r_y H(\Omega)$ and the types of $H(\Omega)$.

**Check C4.** The orbit parity space has dimension 15.

**Check C5.** Every orbit has a verified pure 3-cycle.

**Check C6.** Every generator has zero $C_3$ twist sum on each relevant orbit and even $D_4$ flip parity.

**Check C7.** All 744 applicable sites admit each nonidentity single-cell turn in their rotation subgroup.

**Check C8.** The words $P_\Omega$ and $F$ have the stated actions.

**Check C9.** Verify the group order and move-count lower bounds.

## 8 October 10, 2026 revision

The October 4 draft left two gaps. For orientations it computed the single-cell freedom groups, then assumed a descending move would always be available. The new argument establishes the joint constraints and the orientation kernel, and implements a terminating completion phase. For positions the draft inferred the required pure conjugates directly from the induced alternating groups. The revised proof verifies orbit-local 3-cycles separately. It stores the derived cycles for the 31 orbits and falls back to exact search when greedy progress stops. The timing and move counts above were measured on the earlier implementation.

## 9 Relationship to Level 2

The companion document `level2-completeness.tex` characterizes reachable Level 2 states and gives the group order using corresponding parity and orientation invariants, pure cycles, and two-site tools. For Level 2, transitivity on ordered triples is checked by exhaustive search. For Level 3, primitivity and Jordan's theorem give transitivity through the alternating group.